

Chapter 2

Topologies of DC-DC Converters

The converter is a device, which converts one level of electrical energy into another level of electrical energy efficiently. Generally, in case of DC-DC converters, the energy conversion takes place between two different DC levels. Power supplies are designed to meet some basic needs such as regulated output voltage, isolation and multiple outputs. Other prime goals of converters are to reduce their size and weight and improve their efficiency. Switch mode converters having high energy efficiency ($> 80\%$) compared to linear converter ($30 - 60\%$)[17] are mostly used for majority power supply applications. Figure. 2.1 shows the basic circuit of a linear converter where the switch is always in on condition. The transistor (here the [Bipolar Junction Transistor \(BJT\)](#)) in linear supplies, acts like an adjustable resistor causes a continuous power loss. In case of switched power supplies (in Fig. 2.2), the transistor is either on or off, hence, mainly switching loss is occurred. The following section discusses various switching converter topologies and their operations. All switching converters can be broadly classified into two categories i.e. non-isolated and isolated. Working topologies wise, they are grouped into three basic circuits[17, 37]:

- Buck Converter
- Boost Converter
- Buck-Boost Converter

The operations of each converter are also divided into[9, 17, 37]:

- a) [Continuous Conduction Mode \(CCM\)](#) and
- b) [Discontinuous Conduction Mode \(DCM\)](#)

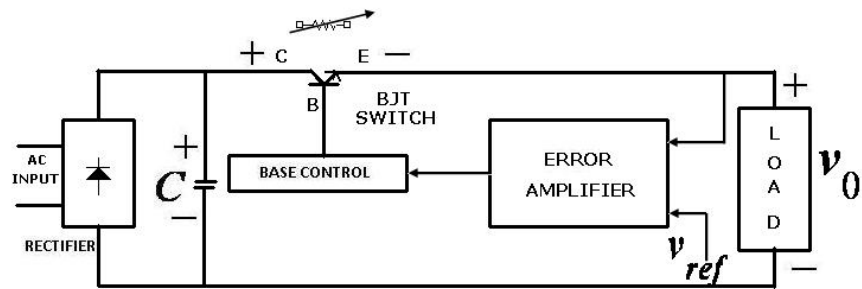


Figure 2.1: Linear converter

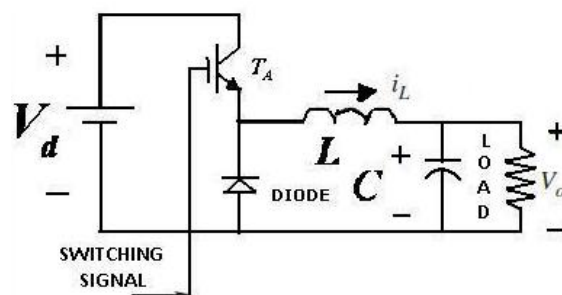


Figure 2.2: Switched mode converter

2.1 Buck Converter

As the name implies, a Buck converter produces a lower average output voltage than the input DC voltage. The Buck converter is generally worked in CCM (inductor current never becomes zero in the switching cycle). The following figure (Fig. 2.3) shows a basic circuit of a Buck converter. Here the transistor is a switch which operates at very high frequency (20kHz-200kHz)[17] to produce chopped voltage (voltage across the diode). This chopped voltage is then filtered by the load side LC 2^{nd} order circuit to get a smooth output voltage (V_0).

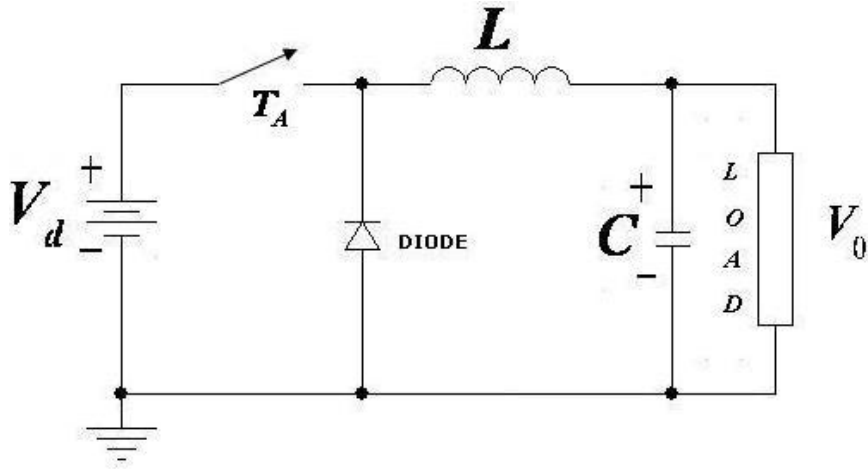


Figure 2.3: Buck converter

- **During switch on:**

During the interval when the switch is on, the load and the inductor are fed from the input as the diode in Fig. 2.4a, becomes reverse biased.

- **During switch off:**

During this interval(in Fig. 2.4b), polarity of the inductor becomes reversed hence the diode is in forward biased. So some of stored energy of the inductor is transferred to the load and filter capacitor.

Here the output filter capacitor is assumed to be large enough so that the instantaneous output voltage is nearly constant (i.e. $v_0(t) \simeq V_0$). Again it is observed that the average inductor current(I_L) is equal to the average load current (I_0), since the average capacitor current in steady state is zero.

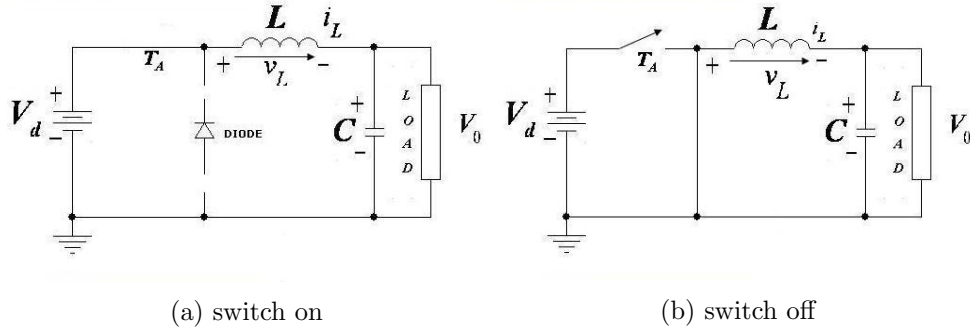


Figure 2.4: Buck converter circuit states (in CCM)

2.1.1 Continuous Conduction Mode (CCM):

In **CCM**, the inductor current will never be zero in any part of the switching cycle. In this section, the switch, inductor and capacitor are taken as ideal and the operation is taken under the steady state condition. When the switch is on for an interval of t_{ON} , the voltage across the inductor becomes positive (i.e. $v_L = (V_d - V_0) > 0$; as $V_d > V_0$) and inductor current flows from the switch to capacitor and load combination as the diode is in reverse biased shown in Fig. 2.4a, no current will flow through it. When the switch is in off condition, the diode becomes forward biased as the polarity of the inductor becomes reversed (i.e. $v_L = -V_0$) and inductor current flows due to the stored energy of the inductor ($= \frac{1}{2} L i_L^2$), shown in Fig. 2.5

Now, the integral of the inductor voltage(v_L) over one time period is zero under the steady state condition. Therefore,

$$\int_0^{T_s} v_L dt = \int_0^{t_{ON}} v_L dt + \int_{t_{ON+}}^{T_s} v_L dt = 0 \quad (2.1)$$

Where, $T_s = t_{ON} + t_{OFF}$

Form Fig. 2.5 , the above Eq. 2.1 can be written,

$$(V_d - V_0)t_{ON} + (-V_0)t_{OFF} = 0 \quad (2.2)$$

$$(V_d - V_0)t_{ON} + (-V_0)(T_s - t_{ON}) = 0 \quad (2.3)$$

$$\frac{V_0}{V_d} = D \quad (2.4)$$

$$\text{where, Duty}(D) = \frac{t_{ON}}{T_s} \quad (2.5)$$

Again if power losses are neglected, then

$$\text{Input power}(P_d) = \text{Output power}(P_0) \quad (2.6)$$

$$V_d I_d = V_0 I_0 \quad (2.7)$$

where I_d, I_0 are the average input and output load current at steady state respectively.

Using Eqs 2.4 and 2.7,

$$\frac{I_0}{I_d} = \frac{1}{D} \quad (2.8)$$

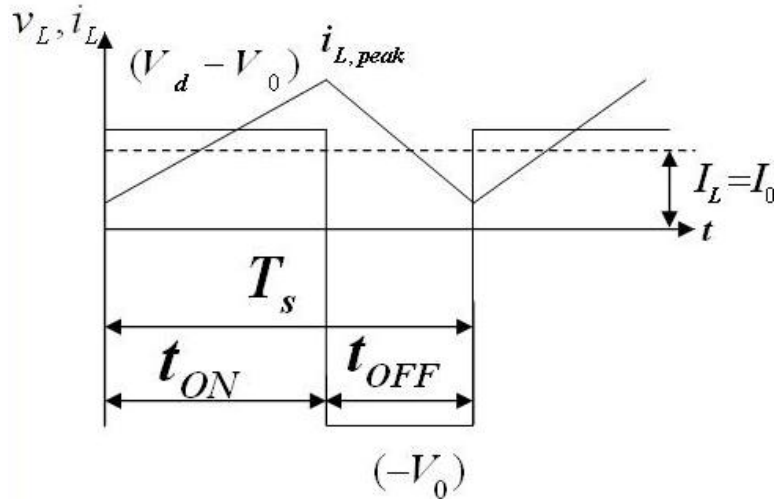


Figure 2.5: Buck converter inductor voltage and current responses (in CCM)

The Eq. 2.4 shows that, in ideal and CCM operating condition, the output voltage of a Buck converter is independent of circuit parameters and linearly varied with input voltage like a DC transformer.

2.1.2 Boundary Condition Between Continuous & Discontinuous Conduction:

At boundary condition, the inductor current (i_L) goes to zero at the end of each switching cycle. Hence from Fig. 2.6 one can write,

$$I_{LB} = \frac{t_{ON}}{2L} (V_d - V_0) \quad (2.9)$$

$$\frac{1}{2} i_{L,peak} = \frac{DT_s}{2L} (V_d - V_0) \quad (2.10)$$

$$= I_{0B} \quad (2.11)$$

Therefore, if the average load current (and, hence, the average inductor current) becomes less than I_{LB} during an operating condition, then i_L becomes discontinuous.

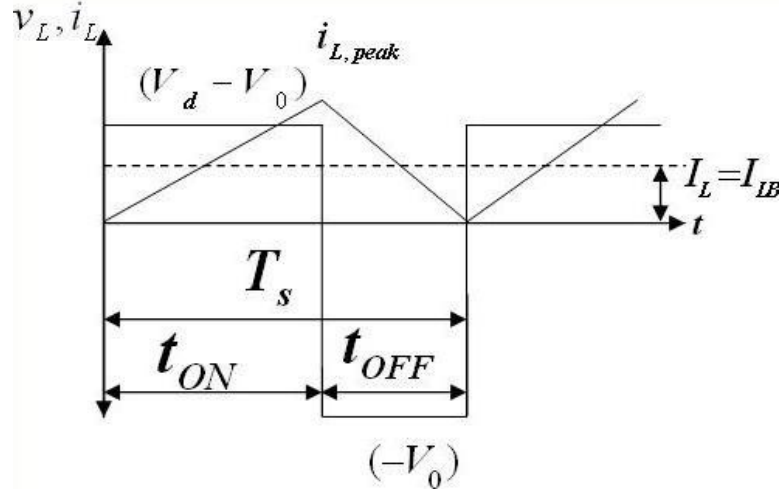


Figure 2.6: Buck Converter inductor voltage and current responses at boundary condition

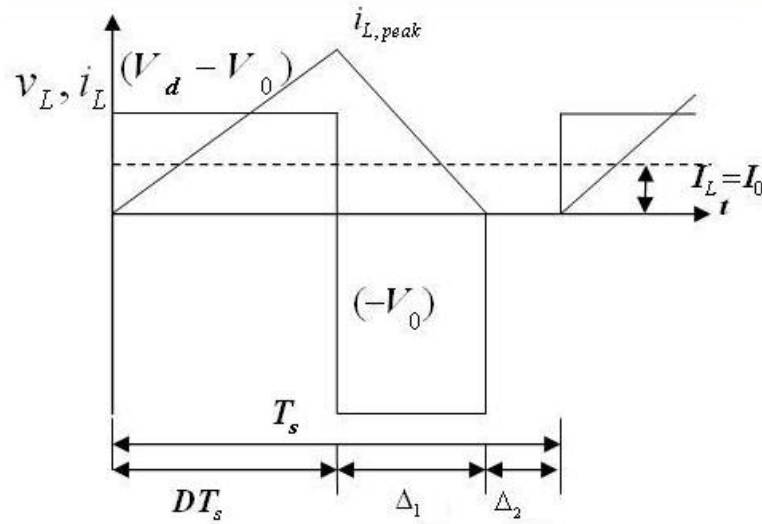


Figure 2.7: Buck converter inductor voltage and current responses (in DCM)

2.1.3 Discontinuous Conduction Mode (DCM):

Many switched mode converters are design to operate in DCM, where for an interval (say Δ_2) of each switching period the i_L remains zero.

If the load on a Buck converter is gradually reduced then after certain reduction of load it will operate in DCM shown in Fig. 2.7

2.2 Boost Converter

As the name suggests, the output voltage is always greater (i.e. boosted up) than the input voltage in case of a Boost Converter[9, 37]. This converter (shown in Fig. 2.8) contains a diode, a transistor as switch and at least one energy storage element. Generally, a capacitor is added to the output side to remove output voltage ripple and sometimes an inductor is combined with it to minimize the current ripple.

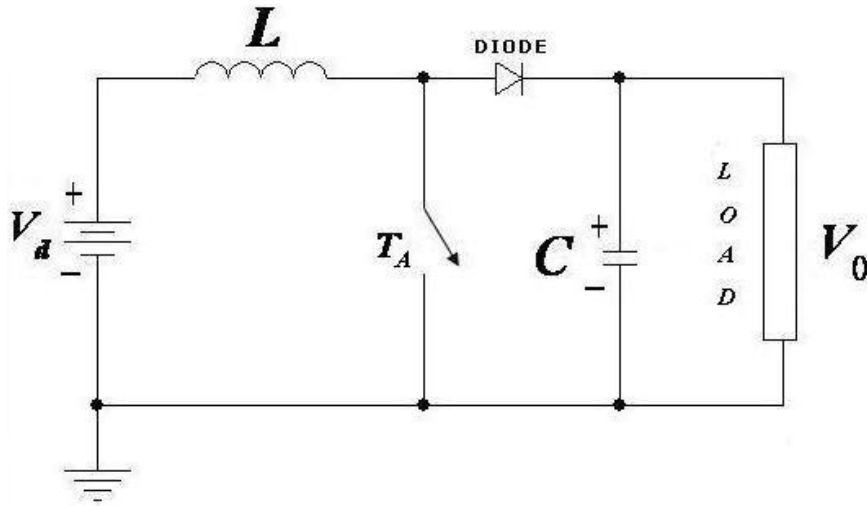


Figure 2.8: Boost converter

- **During switch on:**

In Fig. 2.9a switch is made to close its contacts, which results in increase of inductor current. There no power is transferred from input side to output side as the transistor on time resistance (R_{ON}) is less than diode forward resistance (R_{OND}).

- **During switch off:**

During this interval (in Fig. 2.9b), polarity of the inductor becomes reversed and hence its voltage adds to the input voltage (V_d), making the output voltage greater than the input voltage. The inductor stored energy is transferring to the load. Switch is made to open and thus the only path for inductor current to flow is through the diode and the parallel combination of capacitor and load. This enables capacitor to

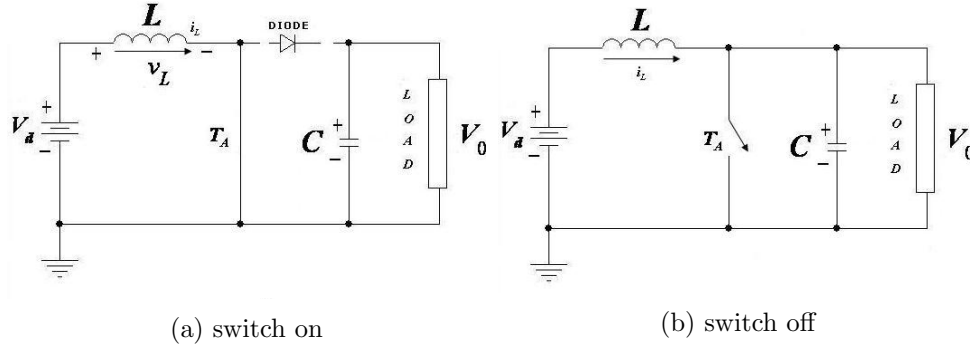


Figure 2.9: Boost converter circuit states (in CCM)

be recharged. Here the output filter capacitor is assumed to be large enough so that the instantaneous output voltage is nearly constant (i.e. $v_0(t) \simeq V_0$).

2.2.1 Continuous Conduction Mode (CCM):

In steady state, the time integral of inductor voltage over one switching period is zero. Therefore from Fig. 2.10,

$$V_d t_{ON} + (V_d - V_0) t_{OFF} = 0 \quad (2.12)$$

Dividing both sides of the Eq. 2.12 by T_s yields

$$\frac{V_0}{V_d} = \frac{1}{1 - D} \quad (2.13)$$

If circuit is assumed to be lossless

$$\text{Input power}(P_d) = \text{Output power}(P_0) \quad (2.14)$$

$$V_d I_d = V_0 I_0 \quad (2.15)$$

Now from Eq. 2.13,

$$\frac{I_0}{I_d} = (1 - D) \quad (2.16)$$

2.2.2 Boundary Condition Between Continuous & Discontinuous Conduction:

At the boundary condition, i_L becomes zero at the end of each switching period as shown in Fig. 2.11. Thus average value of the inductor current at

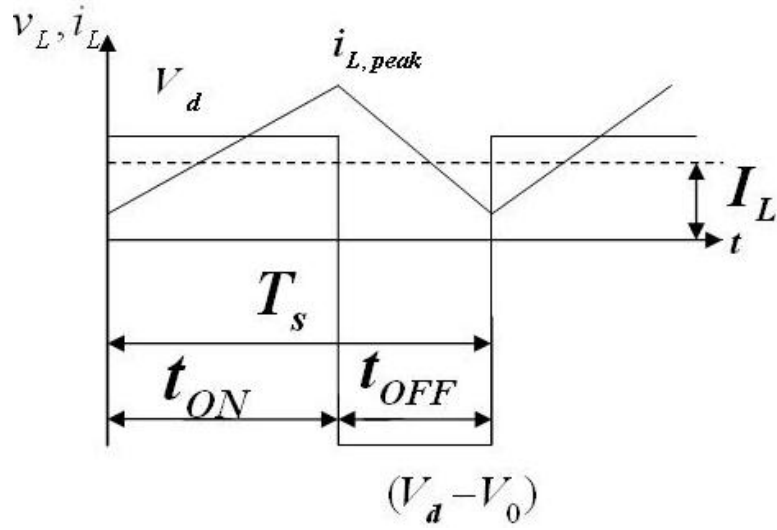


Figure 2.10: Boost converter inductor voltage and current responses (in CCM)

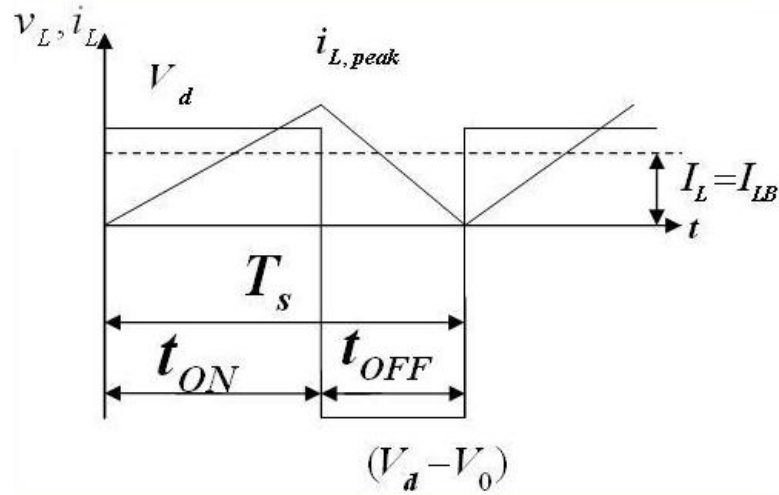


Figure 2.11: Boost converter inductor voltage and current responses at boundary condition

this mode is

$$I_{LB} = \frac{1}{2} i_{L,peak} \quad (2.17)$$

$$= \frac{V_d}{2L} t_{ON} \quad (2.18)$$

$$= \frac{T_s V_d}{2L} D(1 - D) \quad (2.19)$$

From Fig. 2.8, it is noticed that for a Boost converter the inductor current (I_L) and input current (I_d) are same. In this ends of view and using Eq. 2.16, the average output current at the boundary condition can be written as follows

$$I_{0B} = \frac{T_s V_0}{2L} D(1 - D)^2 \quad (2.20)$$

The above Eq. 2.20 shows that for a given duty (D) and constant output voltage (V_0), if the average load current (I_0) goes below than I_{0B} then the converter will run in DCM.

2.2.3 Discontinuous Conduction Mode (DCM):

Again, from Fig. 2.12, the inductor volt-second relationship we can write,

$$V_d t_{ON} + (V_d - V_0) \Delta_1 = 0 \quad (2.21)$$

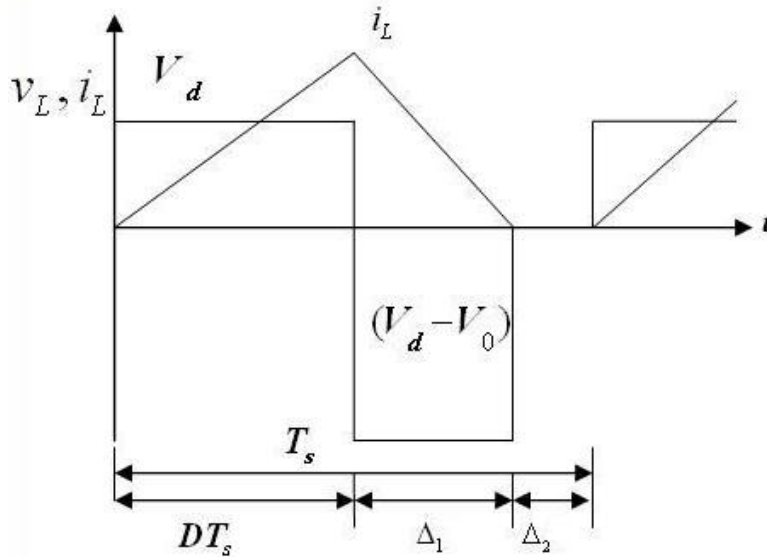


Figure 2.12: Boost converter inductor voltage and current responses (in DCM)

Therefore,

$$V_d D T_s + (V_d - V_0 D_D T_s) = 0 \quad (2.22)$$

$$\frac{V_0}{V_d} = \frac{D_D + D}{D_D} \quad (2.23)$$

Where,

$$D_D = \frac{\Delta_1}{T_s}$$

If power loss is neglected then,

$$\frac{I_0}{I_d} = \frac{D_D}{D_D + D} \quad (2.24)$$

Generally, a Boost converter is preferred to operate in **DCM** rather than **CCM** and the reasons behind this will be discussed in the next chapter.

2.3 Buck-Boost Converter

As the name indicates, a Buck-Boost converter (in Fig. 2.13) produces higher or lower voltage than input side as per duty ratio of the transistor switch of the converter[17, 37]. Another noticeable characteristic of this converter is that the output voltage polarity is reversed with respect to the input polarity.

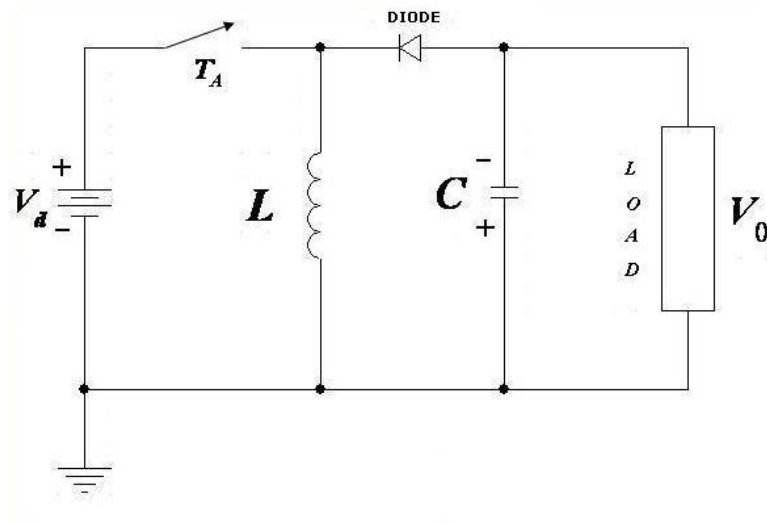


Figure 2.13: Boost converter

- **During switch on:**

From Fig. 2.14a it is clear that when the switch is closed, input provides energy to the inductor. As the diode is in reversed biased hence, no power will flow to the smoothing capacitor as well as to the load from input during this interval. The load is fed by capacitor alone.

- **During switch off:**

In Fig. 2.14b, during this interval, polarity of the inductor becomes reversed and stored energy of the inductor is transferred to the smoothing capacitor and load. Here it is assumed that the output capacitor is large enough to produce constant output voltage (i.e. $v_0(t) \simeq V_0$).

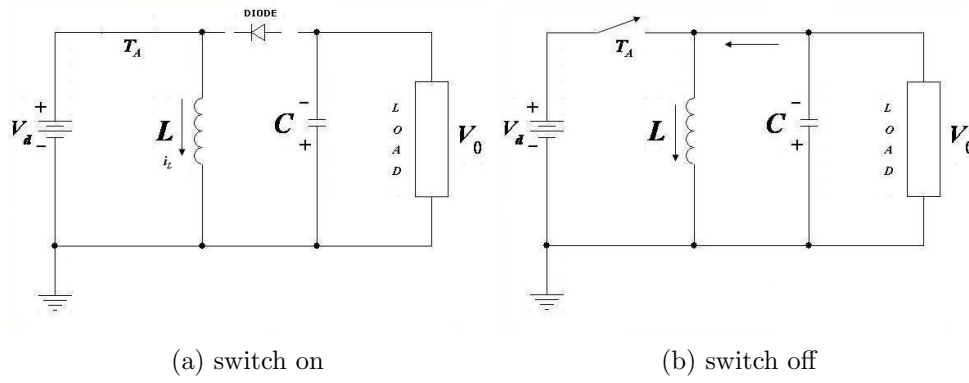


Figure 2.14: Buck-Boost converter circuit states (in CCM)

2.3.1 Continuous Conduction Mode (CCM):

Fig. 2.15 shows in continuous current conduction mode, inductor current will flow continuously.

Now the average voltage across the inductor over each cycle is zero yields

$$V_d t_{ON} + (-V_0) t_{OFF} = 0 \quad (2.25)$$

$$V_d D T_s + (-V_0)(1 - D) T_s = 0 \quad (2.26)$$

$$\frac{V_0}{V_d} = \frac{D}{(1 - D)} \quad (2.27)$$

If power losses are neglected (i.e. $P_d = P_0$) then

$$\frac{I_0}{I_d} = \frac{1 - D}{D} \quad (2.28)$$

The Eq. 2.27 shows that by varying D , the output voltage can be either higher or lower than that of the input.

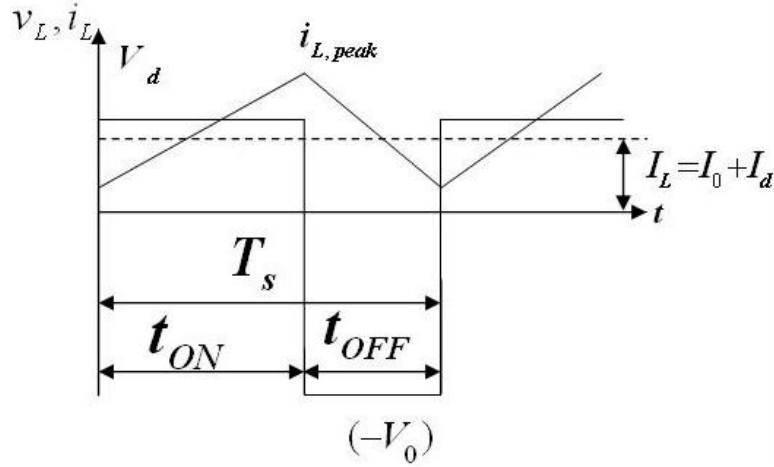


Figure 2.15: Buck-Boost converter inductor voltage and current responses (in CCM)

2.3.2 Boundary Condition Between Continuous & Discontinuous Conduction:

At boundary condition, i_L goes to zero at the end of each switching cycle. From Fig. 2.16

$$I_{LB} = \frac{i_{L,peak}}{2} \quad (2.29)$$

$$= \frac{V_d}{2L} t_{ON} \quad (2.30)$$

Since average capacitor current in steady state is zero. Hence,

$$I_0 = I_L - I_d \quad (2.31)$$

Again using Eqs. 2.27 through 2.31, the average output current at the boundary condition can be written in terms of output voltage (V_0) as follows

$$I_{0B} = \frac{V_0 T_s}{2L} (1 - D)^2 \quad (2.32)$$

If the average load current (I_0) is less than I_{0B} for a particular D and V_0 then it will run under discontinuous conduction mode.

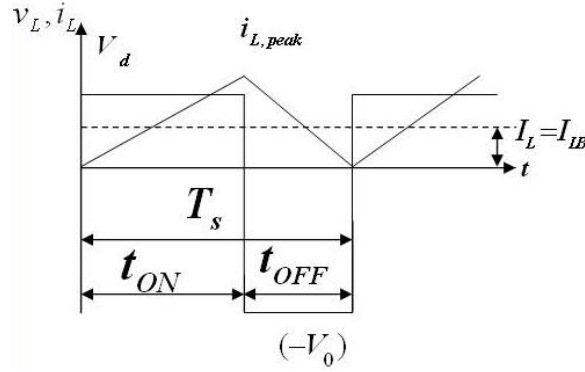


Figure 2.16: Buck-Boost converter inductor voltage and current responses at boundary condition

2.3.3 Discontinuous Conduction Mode (DCM):

In DCM (Fig. 2.17), we can write the following equation from the concept of average voltage across the inductor over one complete cycle is zero :

$$V_d t_{ON} + (-V_0) \Delta_1 = 0 \quad (2.33)$$

$$V_d D T_s + (-V_0) D_D T_s = 0 \quad (2.34)$$

$$\frac{V_0}{V_d} = \frac{D}{D_D} \quad (2.35)$$

Like Boost converter a Buck-Boost converter is also preferred to run in DCM and the reason will be discussed in the next chapter.

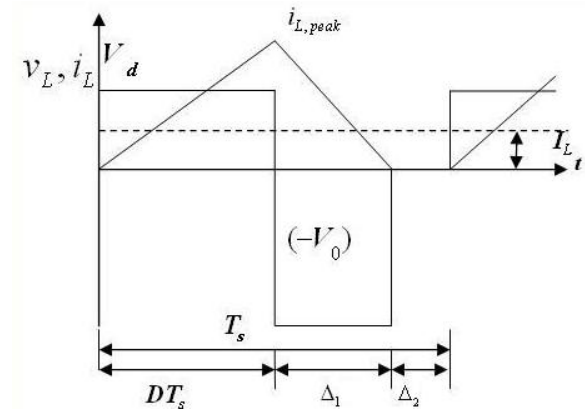


Figure 2.17: Buck-Boost converter inductor voltage and current responses (in DCM)

2.4 Comparison Between CCM and DCM:

In this section, we will try to focus on comparison between continuous and discontinuous mode of operations for different DC-DC converters[37].

A) Continuous Conduction Mode Of Operation:

In CCM, inductor current never falls to zero during any part of the switching cycle. To achieve it, inductance has to be taken larger than boundary value. Generally, inductance is taken three times more than boundary value for very lightly loaded condition. Otherwise, it will have to be taken ten times more than that of the boundary value[16]. It is observed that in CCM, a Boost and a Buck-Boost converters current in output stage (i.e. the diode current) is discontinuous though the inductor current is continuous. However, in the Buck converter, the inductor current is the output stage current and continuous also. Hence it has less ripple compare to other converters and filter design for Buck converter is comparability easier than that of the other converters.

a. Open Loop Regulation :

As in ideal case, the [Line-To-Output \(LTO\)](#) voltage ratio for each converters are independent of the load (shown in Eqs. 2.4, 2.13 and 2.27). Hence, open loop load regulation is good. However, the output voltage (V_0) depends on input voltage (V_d), therefore, open loop line regulation is poor.

b. Closed Loop Response :

The large inductor along with filter capacitor introduce 2^{nd} order delay in feedback control path, which creates an adverse effect on [Phase Margin \(PM\)](#) as well as on closed loop response.

B) Discontinuous Conduction Mode Of Operation:

In DCM, as the inductor current goes to zero at the end of each cycle, hence inductor current has a high peaky response, which introduces an arduous duty to the switch and the smoothing capacitor, as well as the inductor itself.

a. Open Loop Regulation :

In DCM, the output voltage (V_0) of Buck, Boost and Buck-Boost converter depends on load. Therefore, the open loop load regulation is poor. Again, output voltage depends on input voltage hence open loop line regulation is also poor.

b. Closed Loop Response :

The inductor starts each cycle with a zero stored energy. Hence, output current is cycle-by-cycle basis. Therefore, the inductor

current has no effect on small signal response. It is observed that the non-minimum phase response of Boost and Buck-Boost converters will not come into modelling in DCM. Hence, the converters are stable with a good closed loop response.

2.5 Isolated Switched Mode Power Supply (SMPS)

So far what are discussed here, all converters are non-isolated. This “isolation” refers to the existence of an electrical barrier between the input and output of the DC-DC converter. Generally, most DC-DC power supplies are supposed to be isolated. The reasons are as follows:

- The output is electrically isolated from the input high voltage side (230V or 110V) and it also allows the output to be earthed separately, if required[17].
- With the help transformer turns ratio (n), the LTO voltage can be modified because in this case LTO depends not only on duty (D) but also on the turn ratio (n)[37].
- Polarity reversal and multiple outputs can be achieved by choosing appropriate transformer[17].

The isolated DC-DC Converters can be categorised into two main groups[17], as per the way they utilise the transformer core.

A) Unidirectional Core Excitation:

Only the positive part i.e. first quadrant of B-H curve is used (e.g. flyback and forward converters).

B) Bidirectional Core Excitation:

Both parts i.e. first and third quadrants of B-H curve are used alternatively (e.g. push-pull, half and full-bridge converters).

2.5.1 Flyback Converter:

Figure. 2.18 shows that the circuit works like a Buck-Boost converter[17, 37], but here inductor has a secondary winding (or windings for multiple outputs). As per requirement, the transformer should store energy. Hence, the value of magnetizing inductance(L_μ) of this coupled inductor is designed to be as small as possible to store required energy ($\frac{1}{2}Li_L^2$).

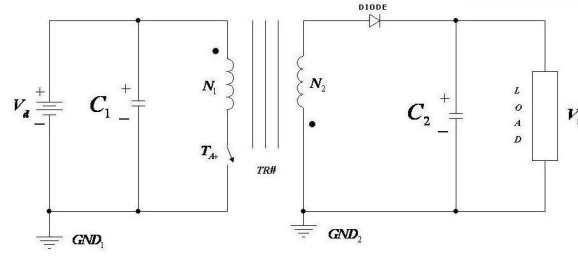


Figure 2.18: Flyback converter

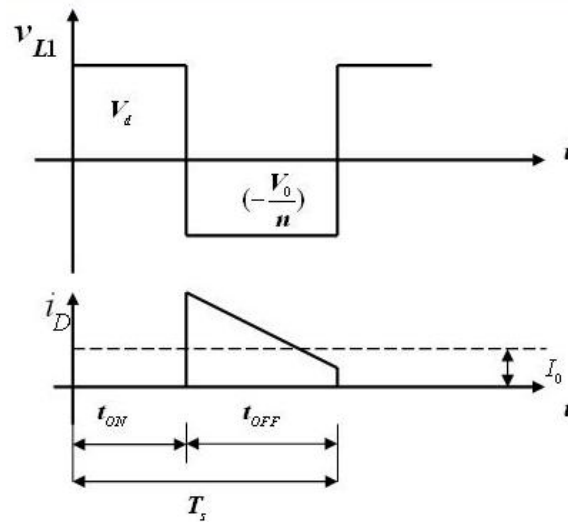


Figure 2.19: Flyback converter primary voltage and diode current waveform

- **During switch on:**

When the switch is on, energy is transferred from the input to the primary of the transformer. As per winding polarity, the diode is reversed biased, hence no current will flow from secondary to the load and capacitor combination through the diode. The load is fed from the filter capacitor alone.

- **During switch off:**

Now the diode becomes forward biased and hence, inductor stored energy flows to capacitor and load combination through the diode. Therefore, it is clear that transformer ratio will come into the relationship of **LTO** voltage.

In **CCM**, **LTO** voltage relationship can be written from Fig. 2.19 as

$$\frac{V_0}{V_d} = \frac{nD}{(1-D)} \quad (2.36)$$

Where $n(= \frac{N_2}{N_1})$ is the turns ratio of the transformer.

2.5.2 Forward Converter:

The forward converter (in Fig. 2.20) is very similar to a Buck converter[17, 37], but the addition of a transformer provides both electrical isolation and the possibility of a wide voltage range.

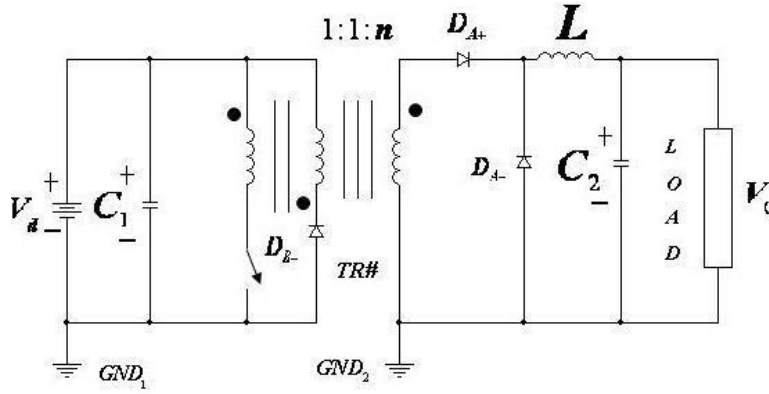


Figure 2.20: Forward converter

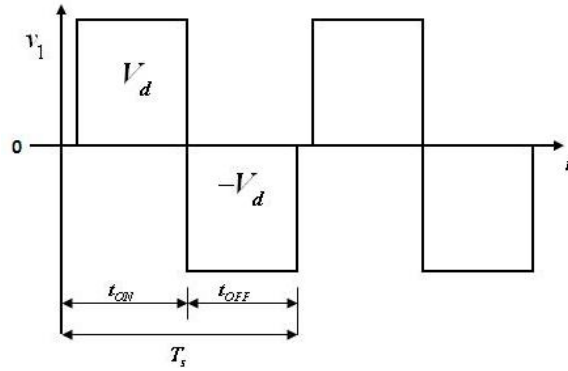


Figure 2.21: Forward converter primary voltage waveform

- **During switch on:**

During this interval, the diode (D_{A+}) is forward biased, and energy is transferred from input to the load capacitor combination.

• **During switch off:**

When the switch is off, diode (D_{A+}) becomes reverse biased and diode (D_{A-}) is forward biased to maintain continuous current in the output circuit.

To avoid core saturation of the transformer a tertiary winding is added[37]. The primary voltage waveform is shown in Fig. 2.21. Therefore, LTO voltage relationship can be written as follows

$$\frac{V_0}{V_d} = nD \quad (2.37)$$

2.5.3 Half-Bridge Converter:

In the half-bridge converter, capacitors C_1 and C_2 form a voltage divider for input voltage (V_d), the average voltage at the centre-point is ($\frac{V_d}{2}$) (in Fig. 2.22). The switches T_{A+} and T_{A-} conduct alternatively, switching the primary voltage from ($+\frac{V_d}{2}$) to ($-\frac{V_d}{2}$). Thus the peak switching voltage is (V_d).

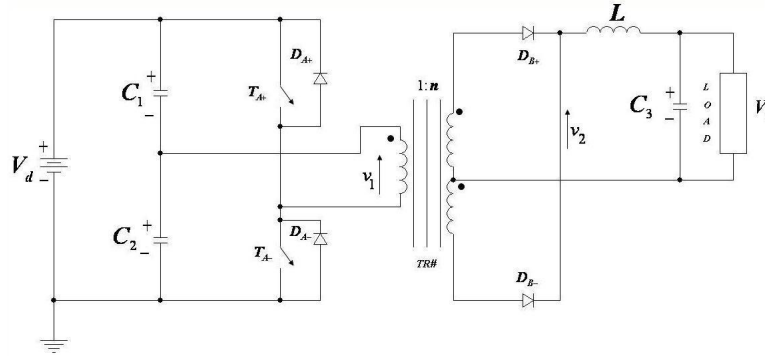


Figure 2.22: half-bridge converter (Isolated)

Therefore, LTO voltage ratio becomes

$$\frac{V_0}{V_d} = nD \quad (2.38)$$

It should be noted that D must be limited to 0.5 to avoid short circuit between switches. Again capacitors (C_1 and C_2) block any DC component to the transformer primary. Hence, the risk of core saturation of the transformer is reduced.

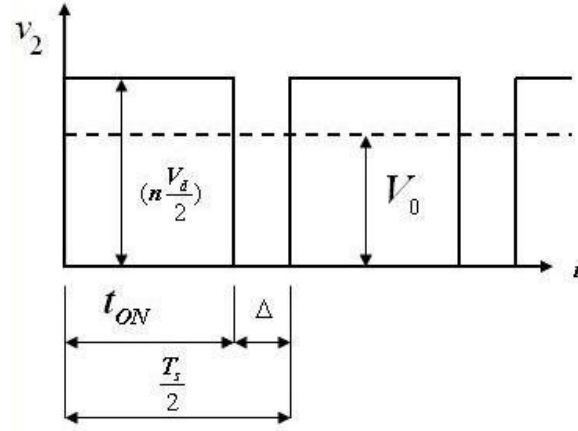


Figure 2.23: Half-bridge converter waveform

2.5.4 Full-Bridge Converter:

In the full-bridge converter (in Figs. 2.24 and 2.25), the switches T_{A+} and T_{B-} are closed simultaneously whereas during this interval switches T_{A-} and T_{B+} remain open. Thus a square AC voltage is produced at the transformer primary, having a waveform equal to $\pm V_d$.

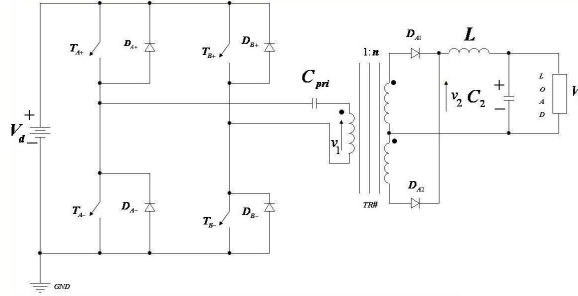


Figure 2.24: Full-bridge converter (Isolated)

Therefore,

$$\frac{V_0}{V_d} = 2nD \quad (2.39)$$

A capacitor (C_{pri}) may be placed to avoid core saturation[37].

2.5.5 Push-pull Converter:

The push-pull converter is a Buck type converter with isolation, shown in Fig. 2.26. The two switches (i.e. T_{A+} and T_{B+}) are closed alternatively.

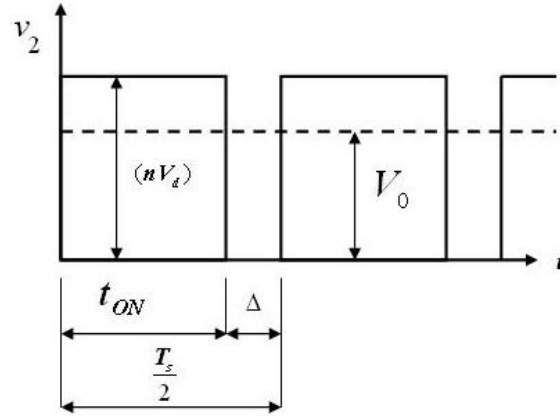


Figure 2.25: Full-bridge converter waveform

The duty ratio (D) of the transistors must be within 50% to prevent “shoot-through” current of the DC supply and care must be taken to ensure that the switches should have equal duty ratio to prevent transformer core saturation.

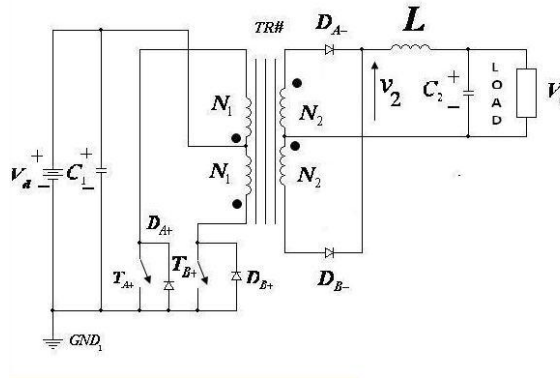


Figure 2.26: Push-pull converter

The LTO relationship of voltage can be written from Fig. 2.27 as

$$\frac{V_0}{V_d} = 2nD \quad (2.40)$$

Where,

D is the duty ratio of each transistor and n is the transformer turns ratio. Here both transistors are referred to the same reference voltage unlike half-bridge or full bridge converters, so they can be driven directly by the same control circuit, without isolating transformer or opto-coupler in the

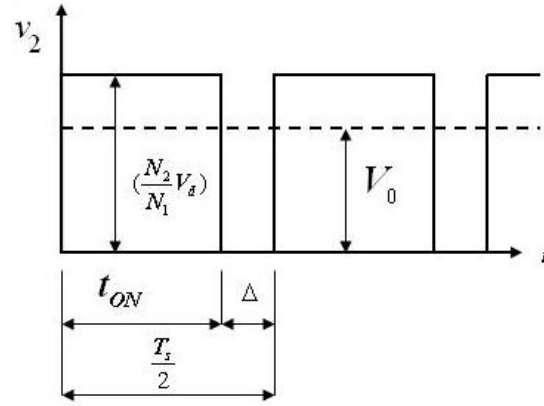


Figure 2.27: Push-pull converter waveform

gate driver circuit.

2.6 Some Other DC-DC Converters:

2.6.1 Cúk Converter:

The basic Cúk¹ converter (in Fig. 2.28) has combination characteristics of low input current ripple of the Boost converter with low output current ripple of the Buck converter[9].

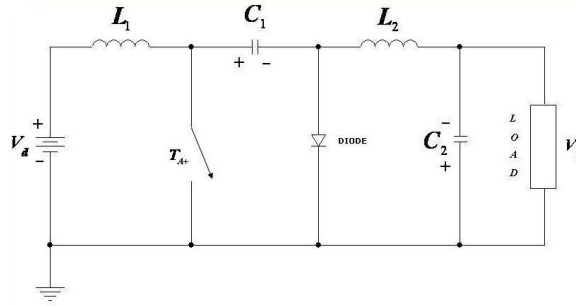


Figure 2.28: Cuk converter

The LTO voltage relationship as follows

$$\frac{V_0}{V_d} = \frac{D}{(1-D)} \quad (2.41)$$

¹named after its inventor Professor Slobodan Cúk of California Institute of Technology

Where V_0 is of the opposite polarity to V_d , as in case of a Buck-Boost converter. The advantages of Cúk converter over Buck-Boost converter are : input and output current ripples are less in this case.

2.6.2 Single Ended Primary Inductance Converter (SEPIC):

The SEPIC (in Fig. 2.29) is a DC-DC converter topology that provides a positive regulated output voltage to be greater than, less than, or equal to its input voltage. A SEPIC is similar to a traditional Buck-Boost converter[37], but has advantages of having non-inverted output (the output has the same voltage polarity as the input).

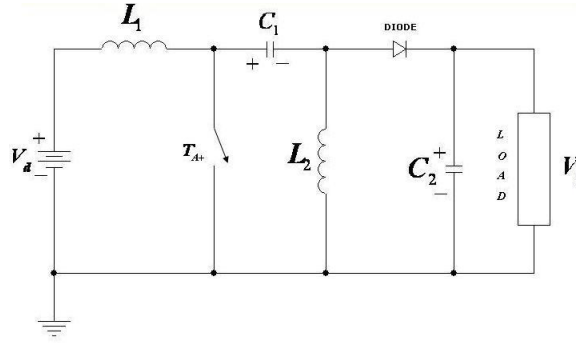


Figure 2.29: SEPIC

Chapter 3

Mathematical Model Of DC-DC Converters

This chapter demonstrates the [State-Space Averaging \(SSA\)](#) modelling of basic DC-DC converter topologies. Power electronic converters are periodic time-variant, non-linear systems, because of their switching operation. The [State-Space Averaging \(SSA\)](#) method is a tool to model them as time independent linear systems, defined by a unified set of differential equations, capable of representing circuit waveforms. Therefore, it can be a convenient approach for designing controllers to be applied to switched converters.

3.1 Techniques To Find Out Converter Models

There are a lot of techniques to figure out the DC-DC converter models such as:

3.1.1 [Circuit Averaging Technique \(CAT\)](#):

Circuit averaging [22] is another well-known technique for derivation of converter equivalent circuits. Rather than averaging the converter state equations, with the circuit averaging technique we average the converter waveforms directly. All manipulations are performed on the circuit diagram, instead of on its equations, and hence the circuit averaging technique gives a more physical interpretation to the model. Since circuit averaging involves averaging and small-signal linearisation, it is equivalent to state-space averaging. However, in many cases circuit averaging is easier to apply, and allows the small-signal ac model to be written almost by inspection.

3.1.1.1 Comparison between CAT and SSA:

The SSA approach has a number of advantages over CAT method as follows:

- More compact representation of equations.
- Ability to obtain more transfer functions than is possible using circuit averaging technique
- Both DC and AC transfer functions are obtained with more eases.

3.1.2 PWM Switch Model (PWMSM) Technique:

This model leads to considerable simplification in the analysis of the non-linear DC-DC converters, by which DC and small-signal AC analysis can be performed by simple circuit analysis.

3.1.2.1 Comparison between PWMSM and SSA:

The main difference between the state space averaging and the [PWM Switch Model](#), is that in the former method, the state equations describing the total system are linearised, while in the [PWM Switch Model](#) only the non-linear part responsible for the switching is linearised. Both techniques use averaging of the quantities of interest, hence they are both accurate for frequencies well below half of the switching frequency [18, 35].

3.2 State-Space Averaging (SSA) Technique

The goal of the following analysis is to find out a small signal transfer function $\frac{\tilde{v}_0(s)}{\tilde{d}(s)}$, where $\tilde{v}_0$ and $\tilde{d}$ are small perturbation in the output voltage (v_0) and the duty ratio (d) respectively, around their steady state operating value V_0 and D respectively. Here all calculations are done for [Continuous Conduction Mode \(CCM\)](#) of operation. In case of [Discontinuous Conduction Mode \(DCM\)](#), the converter has three circuit states where as in CCM there are two circuit states of operations.

The primary result of [SSA](#) is the weighted average of the corresponding matrices of state-space equations in each individual switching mode in one switching cycle, under the basic assumptions [15]:

- 1) The switching frequency (f_s) is much higher than the highest natural frequency (f_0) of the converters in each switching phase.